

JAMES BAILIE

Curriculum Vitae

September 2026

CONTACT INFORMATION

CSE/DSAI
Gibraltarvällsvägen 7
412 58 Göteborg
Sweden

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CURRENT POSITION

Chalmers University of Technology

9/2025–

Doktor (i.e., postdoc)

Division of Data Science and AI, Department of Computer Science and Engineering

Supervisor: Prof. Adel Daoud (AI & Global Development Lab)

PREVIOUS POSITIONS

Institute for Employment Research (IAB)

6/2025–8/2025

Wissenschaftlicher Mitarbeiter (i.e., research fellow)

Supervisor: Prof. Jörg Drechsler

Australian Bureau of Statistics

Mathematical Statistics Specialist

11/2018–8/2020

Graduate

2/2018–11/2018

Methodology Division

Australian Department of the Prime Minister and Cabinet

4/2020–5/2020

On secondment in the COVID-19 Taskforce

Menzies School of Health Research

Data Analyst

7/2013–5/2016

IT Assistant

11/2011–6/2013

EDUCATION

Harvard University

PhD, Statistics, 2025.

Thesis: *Topics in Privacy, Data Privacy and Differential Privacy*

Committee: Profs. Xiao-Li Meng (chair), Kosuke Imai, Salil Vadhan & Adam Smith

MA, Statistics, 2024.

Australian National University

BSc (with first-class honours), Mathematics, 2017.

Thesis: *Vector Fields on Spheres*

Thesis Adviser: Dr. Vigleik Angeltveit

PAPERS

[†] indicates alphabetical ordering of authors and * equal contribution.

PREPRINTS & WORKING PAPERS

15. Markus B. Pettersson, JB, Mohammad Kakooei, Eagon Meng and Adel Daoud (2026). “Beyond Point Predictions: Uncertainty-Aware Satellite Poverty Mapping for Public Policy”. DOI: 10.48550/arXiv.2608.23322
14. JB and Ruobin Gong (2026). “The Five Safes as a Privacy Context”. DOI: 10.48550/arXiv.2510.05803
13. JB[†] and Rabanus Derr[†] (2025). “Property Elicitation on Imprecise Probabilities”. DOI: 10.48550/arXiv.2507.05857
12. Julian Rodemann and JB (2025). “Generalization Bounds and Stopping Rules for Learning with Self-Selected Data”. DOI: 10.48550/arXiv.2505.07367
11. JB and Jörg Drechsler (2024). “Whose Data Is It Anyway? Towards a Formal Treatment of Differential Privacy for Surveys”. *Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches and their Consequences*. Washington, DC: National Bureau of Economic Research. URL: conference.nber.org/conf_papers/f194306.pdf

JOURNAL PUBLICATIONS

10. JB, Ruobin Gong and Xiao-Li Meng (2026). “A Refreshment Stirred, Not Shaken: Invariant-preserving Deployments of Differential Privacy for the U.S. Decennial Census”. *Harvard Data Science Review* (Special Issue 6). ISSN: 2644-2353, DOI: 10.1162/99608f92.dab78690
9. Mohammad Kakooei, JB, Markus B. Pettersson, Albin Söderberg*, Albin Becevic* and Adel Daoud (2026). “A High Resolution Urban and Rural Settlement Map of Africa Using Deep Learning and Satellite Imagery”. *Scientific Reports* 16 (1): 637. ISSN: 2045-2322. DOI: 10.1038/s41598-025-34295-7
8. Matthew J. Schneider, JB and Dawn Iacobucci (2025). “Why Data Anonymization Has Not Taken Off”. *Customer Needs and Solutions* 12 (9). ISSN: 2196-2928. DOI: 10.1007/s40547-025-00158-5
7. JB and Ruobin Gong (2024). “General Inferential Limits under Differential and Pufferfish Privacy”. *International Journal of Approximate Reasoning* 172: 109242. ISSN: 0888-613X. DOI: 10.1016/j.ijar.2024.109242
6. JB (2020). “Big Data, Differential Privacy and National Statistical Organisations”. *Statistical Journal of the IAOS* 36 (4): 1067–1074. ISSN: 1874-7655. DOI: 10.3233/SJI-200685

PEER-REVIEWED CONFERENCE PROCEEDINGS

5. Joshua J. Bon, JB, Judith Rousseau and Christian P. Robert (2026). “Persuasive Privacy”. *Proceedings of the 43rd International Conference on Machine Learning*. Vol. 306. Seoul, South Korea: PMLR. URL: icml.cc/virtual/2026/poster/64335
4. Julian Rodemann, Unai Fischer-Abaigar, JB and Krikamol Muandet (2026). “Performative Learning Theory”. *Proceedings of the 43rd International Conference on Machine Learning*. Vol. 306. Seoul, South Korea: PMLR. URL: icml.cc/virtual/2026/poster/61024
3. JB and Ruobin Gong (2023). “Differential Privacy: General Inferential Limits via Intervals of Measures”. *Proceedings of the Thirteenth International Symposium on Imprecise Probability: Theories and Applications*. Vol. 215. PMLR, pp. 11–24. URL: proceedings.mlr.press/v215/bailie23a.html

PEER-REVIEWED BOOK CHAPTERS

2. Jörg Drechsler and JB (2026). “The Complexities of Differential Privacy for Survey Data”. *Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches, and New Findings*. Ed. by V. Joseph Hotz, Ruobin Gong and Ian M. Schmutte. National Bureau of Economic Research Conference Report. Chicago, IL: University of Chicago Press. ISBN: 978-0-226-85152-5. DOI: 10.7208/chicago/9780226851549.001.0001. URL: nber.org/system/files/chapters/c15024/c15024.pdf
1. JB, Ruobin Gong and Xiao-Li Meng (2026). “Differential Privacy Meets Invariant Statistics: Some Conundrums in Quantifying Trade-Offs”. *Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches, and New Findings*. Ed. by V. Joseph Hotz, Ruobin Gong and Ian M. Schmutte. National Bureau of Economic Research Conference Report. Chicago, IL: University of Chicago Press. ISBN: 978-0-226-85152-5. DOI: 10.7208/chicago/9780226851549.001.0001. URL: nber.org/system/files/chapters/c15017/c15017.pdf

NON-PEER-REVIEWED CONFERENCE PROCEEDINGS

1. JB and Chien-Hung Chien (2019). “ABS Perturbation Methodology through the Lens of Differential Privacy”. *Work Session on Statistical Data Confidentiality, UN Economic Commission for Europe*, p. 13. URL: unece.org/sites/default/files/datastore/fileadmin/DAM/stats/documents/ece/ces/ge.46/2019/mtg1/SDC2019_S2_ABS_Bailie_D.pdf

THESES & EXPOSITORY WORKS

11. JB (2025). “Topics in Privacy, Data Privacy and Differential Privacy”. PhD thesis. Cambridge, MA: Harvard University, p. 465. URL: proquest.com/docview/3217403500/abstract/5B30DA0A3D85414APQ/1
10. JB (2021). “Navigating Spatio-Temporal Data with Generalised Additive Models”. URL: jameshbailie.github.io/files/papers/2021-09-30-Navigating-spatio-temporal-data-with-generalised-additive-models.pdf
9. JB (2021). “Lecture Notes on Asymptotic Statistics”. URL: jameshbailie.github.io/files/papers/2021-06-16-Asymptotic-statistics.pdf
8. JB (2021). “Lecture Notes on Stochastic Processes”. URL: jameshbailie.github.io/files/papers/2021-10-06-Stochastic-processes.pdf
7. JB (2017). “Vector Fields on Spheres”. Honours thesis. Canberra, Australia: Australian National University, p. 112. DOI: 10.25911/5d9efbbf95202
6. JB (2017). “Semisimple Categories”. URL: jameshbailie.github.io/files/papers/2017-06-26-Semisimple-categories.pdf
5. JB (2017). “Abelian Categories and Mitchell’s Embedding Theorem”. URL: jameshbailie.github.io/files/papers/2017-06-16-Abelian-categories-and-Mitchells-embedding-theorem.pdf
4. JB (2017). “A Künneth Formula for Complex K Theory”. URL: jameshbailie.github.io/files/papers/2017-06-06-A-Kunneth-formula-for-complex-K-theory.pdf
3. JB (2017). “Stable Homotopy Theory and Category of Spectra”. Vacation Research Scholarship Report. Australian Mathematical Sciences Institute. URL: jameshbailie.github.io/files/papers/2017-02-01-Stable-homotopy-theory-and-category-of-spectra.pdf
2. JB (2016). “Kolmogorov Complexity and the Symmetry of Algorithmic Information”. URL: jameshbailie.github.io/files/papers/2016-11-01-Kolmogorov-complexity-and-the-symmetry-of-algorithmic-information.pdf
1. JB (2016). “Hausdorff and Similarity Dimensions”. URL: jameshbailie.github.io/files/papers/2016-10-24-Hausdorff-and-similarity-dimensions.pdf

HONOURS

- *Emerging Talent Award* at ISIPTA 2023 for “Differential privacy: General inferential limits via intervals of measures”.
- *Third Place, IAOS Prize for Young Statisticians*, International Association for Official Statistics (2020) for “Big Data, Differential Privacy and National Statistical Organisations”.
- *Fulbright Future Scholarship*, Australian-American Fulbright Commission (2020–2025).
- *2019 Ken Foreman Award*, Australian Bureau of Statistics.
- *Census and Data Services Group Excellence Award*, Australian Bureau of Statistics (2019).
- Undergraduate: *Hanna Neumann Prize for Third Year Mathematics* (2016); *Boyapati Computer Science and Mathematics Prize for Second Year* (2015); *Boyapati Computer Science and Mathematics Prize for First Year* (2014).
- Papers invited (and submitted) to special issues:
 2. JB, Ruobin Gong and Xiao-Li Meng (2026). “A Refreshment Stirred, Not Shaken: Invariant-preserving Deployments of Differential Privacy for the U.S. Decennial Census”. *Harvard Data Science Review* (Special Issue 6). ISSN: 2644-2353, DOI: 10.1162/99608f92.dab78690
 1. JB and Ruobin Gong (2024). “General Inferential Limits under Differential and Pufferfish Privacy”. *International Journal of Approximate Reasoning* 172: 109242. ISSN: 0888-613X. DOI: 10.1016/j.ijar.2024.109242

PROFESSIONAL ACTIVITIES & SERVICE

SERVICE TO THE PROFESSION

- Co-organiser and Discussant, JSM 2026 Topic-Contributed Session, “*Inference from Remote Observations*”.
- Co-organiser, “*Principled Statistics in the Age of AI*” Workshop, Chalmers University, June 2026.
- Communications Chair, Privacy and Confidentiality Interest Group of the American Statistical Association (January 2026–); Communications Committee Member (August–December 2025).
- Co-organiser, 2018 Australian Public Service Graduate Roundtable.
- Co-launcher, Australian Public Service Graduate Data Network, April 2018.
- Journal and Conference Refereeing:
 - 2026: Bayesian Analysis; Journal of Machine Learning Research; 2nd Workshop on Epistemic Intelligence in Machine Learning, ICML 2026 (member of the programme committee); Journal of Official Statistics; Journal of Privacy and Confidentiality (×3).
 - 2025: Journal of Privacy and Confidentiality; International Journal of Approximate Reasoning; Journal of Machine Learning Research; Harvard Data Science Review (×2).
 - 2024: Harvard Data Science Review (×2); Privacy and Public Policy Conference.
 - 2022: Harvard Data Science Review.

UNIVERSITY & DEPARTMENT SERVICE

- Co-organiser, Division of Data Science and AI Seminar Series (June 2026–).
- Panellist, Graduate Student Panel for an Undergraduate Audience, Harvard Statistics Department (October 2024).
- Graduate Student Mentor for first-year statistics PhD students (2022, 2023).
- Member of the Harvard Graduate Statistics Council (2021–2025).
- Co-organiser, Harvard Graduate Statistics Annual Retreat and Faculty Research Lightning Talks (2021–2023).

OTHER ACTIVITIES

- Visitors hosted: Julian Rodemann (LMU, Munich) (September–November 2025).
- Research visits:
 - School of Mathematical Sciences, Adelaide University* (November 2025–January 2026), hosted by Dr. Joshua Bon.
 - Center for Advanced Study in the Behavioral Sciences, Stanford University* (June 2024), hosted by Prof. Adel Daoud.
 - Department of Statistics, Rutgers University* (February 2024), hosted by Prof. Ruobin Gong.
 - Department of Computer Science and Engineering, Chalmers University* (October 2022), hosted by Prof. Adel Daoud.
- Professional associations:
 - International Society of Bayesian Analysis (2024–)
 - Institute of Mathematical Statistics (2022–2025)
 - International Association for Official Statistics (2020–2021)
 - Statistical Society of Australia (2018–2020)

FUNDING

- *Spring 2025 Conference Grant*, US\$750, Harvard Griffin GSAS Student Council.
- *PrivaCI Symposium NSF Travel Grant*, US\$1,000 (2023).
- *Artificial Intelligence Journal Fellowships for PhD Students*, €1,000, ISIPTA 2023.
- *Fulbright Future Scholarship*, ≈US\$299,000, Australian-American Fulbright Commission (2020–2025).
- Undergraduate: *ANU Honours Scholarship*, AU\$5,000 (2017); *Hilary Booth Honours Scholarship*, AU\$5,000 (2017) (declined due to concurrency restrictions); *Vacation Research Scholarship*, ≈AU\$3,000, Australian Mathematical Sciences Institute (2016); *Summer Research Scholarship*, AU\$3,000, University of Queensland (2013–2014).

TEACHING & ADVISING

COURSES TAUGHT AS A TEACHING FELLOW

Harvard University

- STAT 188: *Variations, Information and Privacy* Fall 2023
New course.
Instructor of record: Prof. Xiao-Li Meng
- STAT S-115: *Data Science: An Artificial Ecosystem* Summer 2023
New course.
Instructor of record: Prof. Xiao-Li Meng
- STAT 288: *Deep Statistics: AI and Earth Observations for Sustainable Development*
New graduate course. Spring 2022, 2023
Instructor of record: Prof. Xiao-Li Meng
- STAT 305R: *Statistical Consulting* Fall 2022, Spring 2023
Instructor of record: Prof. Lucas Janson

Australian National University

- MATH 1116: *Advanced Mathematics and Applications 2* Semester 2, 2017
Instructor of record: Dr. Griffith Ware
- MATH 1115: *Advanced Mathematics and Applications 1* Semester 1, 2017
Instructor of record: Dr. Griffith Ware

Other

- NANO 288: *Deep Statistics for More Rigorous and Efficient Data Science* December 2023
2023 Australian Statistical Conference workshop
Instructor of record: Prof. Xiao-Li Meng

UNDERGRADUATE ADVISING

- *Will Hahn* (summer 2024), Harvard College Research Program. Co-advised with Prof. Xiao-Li Meng.
- *Danielle Paulson* (summer 2023), Harvard Program for Research in Science and Engineering, “*A Bayesian Multiple Imputation Procedure to Quantify Uncertainty Due to Privacy Protection*”. Co-advised with Profs. Xiao-Li Meng and Adel Daoud.

SELECTED TALKS

INVITED PRESENTATIONS

30. *Whose Data is It Anyway? Towards a Formal Treatment of Differential Privacy for Surveys, Adelaide Data Privacy Workshop* (keynote), University of Adelaide, Australia, November 2025.
29. *Five Building Blocks of Differential Privacy, Adelaide Data Privacy Workshop* (introductory tutorial), University of Adelaide, Australia, November 2025.
28. *Composition of Privacy Mechanisms: Only Fresh Noise Counts, 9th Bayesian, Fiducial, and Frequentist (BFF9) Conference*, Indianapolis, USA, May 2025.
27. *Differential Privacy in Statistical Agencies—Challenges and Opportunities, 2nd Ocean Workshop on Privacy*, Les Houches, France, February 2025.
26. *How Does Differential Privacy Limit Disclosure Risk? A Precise Prior-To-Posterior Analysis, ISBA World Meeting*, Venice, Italy, July 2024.
25. *Privacy, Data Privacy, and Differential Privacy, Humanising Machine Intelligence Workshop—Celebrating Integrated AI* (keynote, joint with Xiao-Li Meng), Australian National University, Canberra, December 2023.
24. *Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, US Census Bureau Cooperative Agreement Retreat*, Boston University Center for Computing and Data Science, USA, September 2023.
23. *A Discrete Calibration Approach to Improving Data Linkage, ABS Methodology Advisory Committee*, Sydney, Australia, March 2019.

INVITED SEMINARS & COLLOQUIA

22. *Privacy, Data Privacy and Differential Privacy, LMU Munich Statistics Department*, Germany, July 2024.
21. *Privacy, Data Privacy and Differential Privacy, Tübingen AI Center*, Germany, July 2024.
20. *Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, Statistics Canada Methodology Seminar*, virtual, October 2023.
19. *Privacy, Data Privacy and Differential Privacy, Australian Bureau of Statistics*, Canberra, June 2022.

CONFERENCES TALKS

18. *Composition of Privacy Mechanisms: Only Fresh Noise Counts, Joint Statistical Meetings* (topic-contributed paper session), Nashville, USA, August 2025.
17. *Property Elicitation on Imprecise Probabilities, 14th International Symposium on Imprecise Probabilities: Theories and Applications*, Bielefeld, Germany, June 2025.
16. *Enhancing Digital Twins with Privacy-Aware EO-ML Methods, Harvard Center for Geographic Analysis Conference: The Geography of Digital Twins*, Cambridge, USA, May 2025.

15. Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, *Privacy and Public Policy Conference*, Georgetown University, USA, September 2024.
14. Navigating Privacy and Utility with Multiple Imputation, Satellite Imaging and Deep Learning, *Joint Statistical Meetings*, Portland, Oregon, USA, August 2024.
13. Whose Data Is It Anyway? Towards a Formal Treatment of Differential Privacy for Surveys, *Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches and Their Consequences*, NBER, Washington DC, USA, May 2024.
12. Statistical Preprocessing for Privacy-Induced Spatial Mismatch: A Multiple Imputation Approach to Poverty Prediction Using Deep Learning, *Australian Statistical Conference*, Wollongong, Australia, December 2023.
11. Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, *Australian Statistical Conference*, Wollongong, Australia, December 2023.
10. The Five Safes as a Privacy Context, *5th Annual Symposium on Applications of Contextual Integrity*, Toronto, Canada, September 2023.
9. Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, *Joint Statistical Meetings*, Toronto, Canada, August 2023.
8. Differential Privacy: General Inferential Limits via Intervals of Measures, *13th International Symposium on Imprecise Probabilities: Theories and Applications*, Oviedo, Spain, July 2023.
7. Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, *36th New England Statistical Symposium*, Boston University, USA, June 2023.
6. Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, *Data Privacy Protection and the Conduct of Applied Research: Methods, Approaches and Their Consequences*, NBER, Cambridge, USA, May 2023.
5. Designing Formally Private Mechanisms for the $p\%$ Rule, *Workshop on Advances in Statistical Disclosure Limitation*, National Institute for Applied Statistics Research Australia, University of Wollongong, February 2020.
4. Using Admin Data and Machine Learning to Predict Dwelling Occupancy on Census Night, *Statistical Society of Australia's Young Statisticians Conference*, Canberra, October 2019.
3. Stable Homotopy Theory, *Australian Mathematical Sciences Institute Connect Conference*, University of Melbourne, February 2017.

GUEST LECTURES

2. Can Swapping Be Differentially Private? A Refreshment Stirred, Not Shaken, *STAT 188*, *Harvard University*, Cambridge, USA, September 2024.
1. On the Mismatch Between the Prior and the Likelihood Resolutions When Accounting for DHS Privacy Protection, *STAT 288*, *Harvard University*, Cambridge, USA, April 2024.

CODE & DATA

A High Resolution Urban and Rural Settlement Map of Africa Using Deep Learning and Satellite Imagery.

- Open data: [10.7910/DVN/YFRECD](https://doi.org/10.7910/DVN/YFRECD).
- Open-source code: github.com/AlandGlobalDevelopmentLab/Africa-Rural-Urban-Map.

Last updated: 3rd September 2026.